

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Mathematics education plays a crucial role in developing logical reasoning, analytical thinking, and problem-solving abilities among students. In the context of secondary education in Sri Lanka, mathematics is a core subject that significantly influences students' academic and professional pathways. However, despite its importance, many students demonstrate difficulties in understanding mathematical concepts and applying them to solve problems effectively.

One of the major contributing factors to this issue is the teaching approach adopted in classrooms. Traditional teaching methods in mathematics often emphasize procedural knowledge and rote learning rather than conceptual understanding and problem-solving skills. As a result, students may perform well in routine exercises but struggle when faced with unfamiliar or complex problems.

Problem-solving strategies are widely recognized as essential tools in mathematics education. These strategies include understanding the problem, devising a plan, carrying out the plan, and reflecting on the solution—principles popularized by George Pólya. The effective use of such strategies helps learners develop deeper understanding and adaptability in mathematics.

Senior secondary teachers play a vital role in fostering these skills. Their competency in applying problem-solving strategies directly impacts students' learning outcomes. However, there is evidence that many teachers lack adequate training, resources, or confidence to integrate these strategies effectively into their teaching practices.

Therefore, enhancing teachers' competency in using problem-solving strategies is essential to improve the quality of mathematics education in Sri Lanka.

1.2 Statement of the Problem

Despite the recognized importance of problem-solving in mathematics, many senior secondary students in Sri Lanka exhibit low performance in solving non-routine mathematical problems. This issue may stem from teachers' limited use of effective problem-solving strategies during instruction.

Teachers often rely on traditional, teacher-centered methods that focus on memorization and repetitive practice. Consequently, students are not adequately exposed to heuristic approaches or critical thinking processes required for solving complex problems.

The problem addressed in this study is the insufficient competency among senior secondary mathematics teachers in Sri Lanka in utilizing problem-solving strategies effectively in their teaching practices.

1.3 Purpose of the Study

The main purpose of this study is to enhance the competency of senior secondary mathematics teachers in Sri Lanka in using problem-solving strategies in their teaching.

1.4 Objectives of the Study

- I. Identify the current level of cognitive abilities of secondary mathematics teachers regarding problem-solving strategies.
- II. Explore the level of use of problem solving strategies in mathematics by senior secondary teachers.
- III. Find out areas for improvement in problem-solving strategy among senior secondary teachers.
- IV. Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics.
- V. Assess the progress of use of problem-solving strategies in mathematics after intervention.
- VI. Propose a teacher education module on effective use of problem solving strategies in mathematics in senior secondary education.

1.5 Research Questions

The study is guided by the following research questions:

1. What is the current level of competency of senior secondary mathematics teachers in using problem-solving strategies?
2. What challenges do teachers face in implementing these strategies?
3. How can teachers' competency in using problem-solving strategies be enhanced?
4. What is the impact of the intervention on teachers' instructional practices?

1.6 Significance of the Study

This study is significant for several reasons:

- **Teachers:** It will help teachers improve their instructional strategies and professional competence.
- **Students:** Improved teaching methods will enhance students' problem-solving skills and overall achievement in mathematics.
- **Educational Authorities:** Findings can inform curriculum development and teacher training programs in Sri Lanka.
- **Researchers:** The study will contribute to existing literature on mathematics education and teacher development.

1.7 Scope of the Study

This study focuses on senior secondary mathematics teachers in Sri Lanka. It specifically examines their competency in using problem-solving strategies in classroom instruction. The study may be limited to selected schools or regions due to time and resource constraints.

1.8 Limitations of the Study

The study may face the following limitations:

- Limited sample size, which may affect generalizability.
- Time constraints in implementing and evaluating the intervention.
- Possible bias in self-reported data from teachers.
- Variations in school environments and resources.

1.9 Definition of Key Terms

- **Problem-Solving Strategy:** A systematic approach used to understand and solve mathematical problems (e.g., strategies proposed by George Pólya).
- **Competency:** The combination of knowledge, skills, and attitudes required to perform a task effectively.
- **Senior Secondary Teachers:** Teachers who instruct students in upper secondary grades (typically Grades 10–13 in Sri Lanka).
- **Mathematics Achievement:** The level of students' performance in mathematics as measured by assessments.